Homework 1 - Multivariate Analysis

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1. Exploratory Data Analysis

1a. Data Import

```
##         TEAM                 PLAYER POSITION GP GS      Min PTS  X2P. X3P.  FT.
## 1 OLYMPIAKOS          THOMAS WALKUP    Guard 37 37 24:01:00 5.3  47.2 36.7 76.9
## 2 OLYMPIAKOS         FRANK NTILIKNA    Guard 24  6 16:24:00 4.5  40.0 36.5 84.2
## 3 OLYMPIAKOS           KEENAN EVANS    Guard  1  0  2:01:00 0.0   0.0  0.0  0.0
## 4 OLYMPIAKOS             TYSON WARD  Forward 35 14 19:05:00 7.1  56.7 26.4 68.4
## 5 OLYMPIAKOS GIANNOULIS LARENTZAKIS    Guard  7  0  2:59:00 0.3 100.0  0.0  0.0
## 6 OLYMPIAKOS             SABEN LEE     Guard 11  0 11:24:00 3.4  44.4 36.4 81.8
##    OR  DR  TR AST STL  TO BLK BLKA  FC  FD PIR
## 1 0.6 2.0 2.6 5.7 0.9 1.5 0.0  0.1 2.4 1.8 9.5
## 2 0.5 0.6 1.1 1.6 0.5 1.2 0.4  0.2 2.9 1.4 2.6
## 3 0.0 0.0 0.0 0.0 0.0 0.0 0.0  0.0 0.0 0.0 0.0
## 4 0.7 2.2 2.9 2.3 0.5 1.1 0.1  0.3 2.4 2.3 7.8
## 5 0.0 0.1 0.1 0.9 0.0 0.3 0.0  0.0 0.6 0.3 0.0
## 6 0.0 0.4 0.4 1.4 0.4 1.3 0.1  0.2 1.2 0.6 1.8
```

```
## 'data.frame':    71 obs. of  21 variables:
##  $ TEAM    : chr  "OLYMPIAKOS" "OLYMPIAKOS" "OLYMPIAKOS" "OLYMPIAKOS" ...
##  $ PLAYER  : chr  "THOMAS WALKUP" "FRANK NTILIKNA" "KEENAN EVANS" "TYSON WARD" ...
##  $ POSITION: chr  "Guard" "Guard" "Guard" "Forward" ...
##  $ GP      : int  37 24 1 35 7 11 2 8 39 36 ...
##  $ GS      : int  37 6 0 14 0 0 0 0 39 21 ...
##  $ Min     : chr  "24:01:00" "16:24:00" "2:01:00" "19:05:00" ...
##  $ PTS     : num  5.3 4.5 0 7.1 0.3 3.4 0 1.6 19 2.9 ...
##  $ X2P.    : num  47.2 40 0 56.7 100 44.4 0 30 61.8 64.3 ...
##  $ X3P.    : num  36.7 36.5 0 26.4 0 36.4 0 0 40.5 31.1 ...
##  $ FT.     : num  76.9 84.2 0 68.4 0 81.8 0 58.3 89 28.6 ...
##  $ OR      : num  0.6 0.5 0 0.7 0 0 0.5 0.2 1.4 0.3 ...
##  $ DR      : num  2 0.6 0 2.2 0.1 0.4 1 1 5 1.4 ...
##  $ TR      : num  2.6 1.1 0 2.9 0.1 0.4 1.5 1.2 6.4 1.8 ...
##  $ AST     : num  5.7 1.6 0 2.3 0.9 1.4 2.5 1.4 1.3 1.6 ...
##  $ STL     : num  0.9 0.5 0 0.5 0 0.4 0 0.1 0.6 0.4 ...
##  $ TO      : num  1.5 1.2 0 1.1 0.3 1.3 0 0.5 1.1 1 ...
##  $ BLK     : num  0 0.4 0 0.1 0 0.1 1 0.1 0.1 0.1 ...
##  $ BLKA    : num  0.1 0.2 0 0.3 0 0.2 0 0.2 0.2 0.1 ...
##  $ FC      : num  2.4 2.9 0 2.4 0.6 1.2 0 1 0.9 1.7 ...
##  $ FD      : num  1.8 1.4 0 2.3 0.3 0.6 0.5 0.9 3.1 0.3 ...
##  $ PIR     : num  9.5 2.6 0 7.8 0 1.8 5.5 1.2 22.1 2.3 ...
```

```
##         TEAM                 PLAYER POSITION GP GS      Min PTS  X2P. X3P.  FT.
```

```
## 1 OLYMPIAKOS          THOMAS WALKUP    Guard 37 37 24:01:00 5.3  47.2 36.7 76.9
## 2 OLYMPIAKOS         FRANK NTILIKNA    Guard 24  6 16:24:00 4.5  40.0 36.5 84.2
## 3 OLYMPIAKOS           KEENAN EVANS    Guard  1  0  2:01:00 0.0   0.0  0.0  0.0
## 4 OLYMPIAKOS             TYSON WARD  Forward 35 14 19:05:00 7.1  56.7 26.4 68.4
## 5 OLYMPIAKOS GIANNOULIS LARENTZAKIS    Guard  7  0  2:59:00 0.3 100.0  0.0  0.0
## 6 OLYMPIAKOS             SABEN LEE     Guard 11  0 11:24:00 3.4  44.4 36.4 81.8
##    OR  DR  TR AST STL  TO BLK BLKA  FC  FD PIR
## 1 0.6 2.0 2.6 5.7 0.9 1.5 0.0  0.1 2.4 1.8 9.5
## 2 0.5 0.6 1.1 1.6 0.5 1.2 0.4  0.2 2.9 1.4 2.6
## 3 0.0 0.0 0.0 0.0 0.0 0.0 0.0  0.0 0.0 0.0 0.0
## 4 0.7 2.2 2.9 2.3 0.5 1.1 0.1  0.3 2.4 2.3 7.8
## 5 0.0 0.1 0.1 0.9 0.0 0.3 0.0  0.0 0.6 0.3 0.0
## 6 0.0 0.4 0.4 1.4 0.4 1.3 0.1  0.2 1.2 0.6 1.8
```

1b. Convert Minutes

```
##        Min Min 2
## 1 24:01:00    24
## 2 16:24:00    16
## 3  2:01:00     2
## 4 19:05:00    19
## 5  2:59:00     2
## 6 11:24:00    11
```

1c. Set Player Names as Row Labels

```
##                              TEAM POSITION GP GS PTS  X2P. X3P.  FT.  OR  DR
## THOMAS WALKUP          OLYMPIAKOS    Guard 37 37 5.3  47.2 36.7 76.9 0.6 2.0
## FRANK NTILIKNA         OLYMPIAKOS    Guard 24  6 4.5  40.0 36.5 84.2 0.5 0.6
## KEENAN EVANS           OLYMPIAKOS    Guard  1  0 0.0   0.0  0.0  0.0 0.0 0.0
## TYSON WARD             OLYMPIAKOS  Forward 35 14 7.1  56.7 26.4 68.4 0.7 2.2
## GIANNOULIS LARENTZAKIS OLYMPIAKOS    Guard  7  0 0.3 100.0  0.0  0.0 0.0 0.1
## SABEN LEE              OLYMPIAKOS    Guard 11  0 3.4  44.4 36.4 81.8 0.0 0.4
##                           TR AST STL  TO BLK BLKA  FC  FD PIR Min 2
## THOMAS WALKUP          2.6 5.7 0.9 1.5 0.0  0.1 2.4 1.8 9.5    24
## FRANK NTILIKNA         1.1 1.6 0.5 1.2 0.4  0.2 2.9 1.4 2.6    16
## KEENAN EVANS           0.0 0.0 0.0 0.0 0.0  0.0 0.0 0.0 0.0     2
## TYSON WARD             2.9 2.3 0.5 1.1 0.1  0.3 2.4 2.3 7.8    19
## GIANNOULIS LARENTZAKIS 0.1 0.9 0.0 0.3 0.0  0.0 0.6 0.3 0.0     2
## SABEN LEE              0.4 1.4 0.4 1.3 0.1  0.2 1.2 0.6 1.8    11
```

1d. Filter Players

```
## [1] 71
```

```
## [1] 60
```

```
##                      TEAM POSITION GP GS  PTS X2P. X3P.  FT.  OR  DR  TR AST
## THOMAS WALKUP  OLYMPIAKOS    Guard 37 37  5.3 47.2 36.7 76.9 0.6 2.0 2.6 5.7
## FRANK NTILIKNA OLYMPIAKOS    Guard 24  6  4.5 40.0 36.5 84.2 0.5 0.6 1.1 1.6
```

```
## TYSON WARD     OLYMPIAKOS  Forward 35 14  7.1 56.7 26.4 68.4 0.7 2.2 2.9 2.3
## SABEN LEE      OLYMPIAKOS    Guard 11  0  3.4 44.4 36.4 81.8 0.0 0.4 0.4 1.4
## MONTE MORRIS   OLYMPIAKOS    Guard  8  0  1.6 30.0  0.0 58.3 0.2 1.0 1.2 1.4
## SASHA VEZENKOV OLYMPIAKOS  Forward 39 39 19.0 61.8 40.5 89.0 1.4 5.0 6.4 1.3
##                STL  TO BLK BLKA  FC  FD  PIR Min 2
## THOMAS WALKUP  0.9 1.5 0.0  0.1 2.4 1.8  9.5    24
## FRANK NTILIKNA 0.5 1.2 0.4  0.2 2.9 1.4  2.6    16
## TYSON WARD     0.5 1.1 0.1  0.3 2.4 2.3  7.8    19
## SABEN LEE      0.4 1.3 0.1  0.2 1.2 0.6  1.8    11
## MONTE MORRIS   0.1 0.5 0.1  0.2 1.0 0.9  1.2    11
## SASHA VEZENKOV 0.6 1.1 0.1  0.2 0.9 3.1 22.1    27
```

2. Principal Component Analysis (PCA)

2a

```
##
## Call:
## PCA(X = data, scale.unit = TRUE, quanti.sup = quanti_vars, quali.sup = quali_vars,
##      graph = FALSE)
##
##
## Eigenvalues
##                        Dim.1  Dim.2  Dim.3  Dim.4  Dim.5
## Variance               6.681  3.278  1.319  1.222  0.837
## % of var.             39.302 19.285  7.759  7.188  4.924
## Cumulative % of var. 39.302 58.587 66.346 73.534 78.458
##
## Individuals (the 10 first)
##                       Dist    Dim.1    ctr   cos2    Dim.2    ctr   cos2
## THOMAS WALKUP      |  4.651 |  1.845  0.849  0.157 |  2.568  3.352  0.305 |
## FRANK NTILIKNA     |  3.202 | -1.345  0.451  0.176 |  1.163  0.687  0.132 |
## TYSON WARD         |  1.832 |  0.807  0.163  0.194 |  0.342  0.059  0.035 |
## SABEN LEE          |  4.053 | -3.282  2.686  0.656 |  1.753  1.562  0.187 |
## MONTE MORRIS       |  5.268 | -4.171  4.339  0.627 | -0.092  0.004  0.000 |
## SASHA VEZENKOV     |  6.218 |  4.716  5.548  0.575 | -0.355  0.064  0.003 |
## KOSTAS PAPANIKOLAU |  4.201 | -1.676  0.701  0.159 | -0.759  0.293  0.033 |
## TYLER DORSEY       |  5.155 |  3.567  3.173  0.479 |  2.836  4.088  0.303 |
## ALEC PETERS        |  2.435 | -0.118  0.003  0.002 | -0.075  0.003  0.001 |
## NIKOLA MILUTINOV   |  7.115 |  4.382  4.791  0.379 | -3.960  7.974  0.310 |
##                      Dim.3    ctr   cos2
## THOMAS WALKUP        0.739  0.691  0.025 |
## FRANK NTILIKNA      -0.968  1.185  0.091 |
## TYSON WARD          -0.491  0.305  0.072 |
## SABEN LEE           -0.720  0.655  0.032 |
## MONTE MORRIS        -1.859  4.369  0.125 |
## SASHA VEZENKOV       1.489  2.802  0.057 |
## KOSTAS PAPANIKOLAU   1.403  2.489  0.112 |
## TYLER DORSEY        -0.782  0.773  0.023 |
## ALEC PETERS          1.436  2.607  0.348 |
## NIKOLA MILUTINOV    -0.358  0.162  0.003 |
##
```

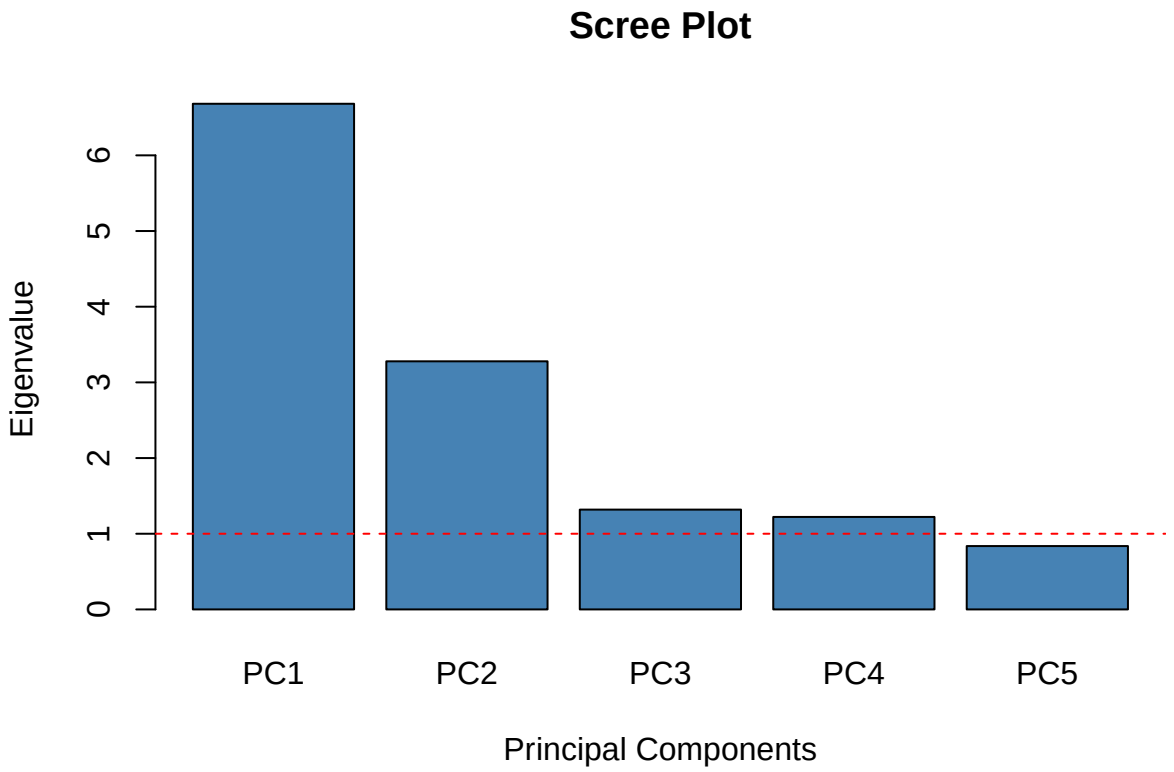
```
## Variables (the 10 first)
##                       Dim.1    ctr   cos2    Dim.2    ctr   cos2    Dim.3
## GP                  |  0.703  7.406  0.495 | -0.022  0.015  0.000 |  0.467
## GS                  |  0.732  8.030  0.537 | -0.001  0.000  0.000 |  0.231
## PTS                 |  0.857 11.002  0.735 |  0.296  2.674  0.088 | -0.022
## X2P.                |  0.481  3.459  0.231 | -0.538  8.844  0.290 |  0.187
## X3P.                |  0.050  0.038  0.003 |  0.561  9.592  0.314 |  0.630
## FT.                 |  0.191  0.544  0.036 |  0.418  5.326  0.175 | -0.083
## OR                  |  0.577  4.990  0.333 | -0.732 16.352  0.536 | -0.062
## DR                  |  0.817  9.989  0.667 | -0.380  4.402  0.144 |  0.122
## TR                  |  0.788  9.300  0.621 | -0.542  8.975  0.294 |  0.063
## AST                 |  0.475  3.383  0.226 |  0.702 15.021  0.492 | -0.091
##                       ctr   cos2
## GP                 16.508  0.218 |
## GS                  4.033  0.053 |
## PTS                 0.037  0.000 |
## X2P.                2.661  0.035 |
## X3P.               30.128  0.397 |
## FT.                 0.527  0.007 |
## OR                  0.290  0.004 |
## DR                  1.133  0.015 |
## TR                  0.305  0.004 |
## AST                 0.631  0.008 |
##
## Supplementary continuous variable
##                       Dim.1   cos2    Dim.2   cos2    Dim.3   cos2
## PIR                 |  0.921  0.849 |  0.054  0.003 | -0.022  0.000 |
##
## Supplementary categories
##                        Dist    Dim.1   cos2 v.test    Dim.2   cos2 v.test
## FENERBAHCE          |  0.844 | -0.573  0.460 -1.070 |  0.222  0.069  0.592 |
## OLYMPIAKOS          |  0.620 |  0.242  0.152  0.414 |  0.054  0.008  0.132 |
## REAL MADRID         |  0.582 |  0.245  0.177  0.402 | -0.239  0.168 -0.559 |
## VALENCIA BASKET     |  0.537 |  0.192  0.128  0.314 | -0.088  0.027 -0.207 |
## Center              |  2.163 | -0.055  0.001 -0.099 | -2.110  0.951 -5.398 |
## Forward             |  1.472 |  1.149  0.609  2.236 |  0.092  0.004  0.254 |
## Guard               |  1.509 | -0.762  0.255 -1.980 |  1.235  0.670  4.582 |
##                      Dim.3   cos2 v.test
## FENERBAHCE          -0.093  0.012 -0.390 |
## OLYMPIAKOS          -0.124  0.040 -0.481 |
## REAL MADRID          0.030  0.003  0.110 |
## VALENCIA BASKET      0.216  0.163  0.798 |
## Center              -0.348  0.026 -1.404 |
## Forward              0.661  0.202  2.895 |
## Guard               -0.244  0.026 -1.425 |
```

2b. Number of Principal Components

```
##        eigenvalue percentage of variance cumulative percentage of variance
## comp 1  6.6813635              39.302138                           39.30214
## comp 2  3.2784135              19.284785                           58.58692
## comp 3  1.3189846               7.758733                           66.34566
## comp 4  1.2219881               7.188165                           73.53382
```

```
## comp 5  0.8371559              4.924447                        78.45827
```

```
## [1] 4
```



Based on the Kaiser criterion, four principal components should be retained, as their eigenvalues exceed 1. The scree plot shows a particularly steep decline after PC2, indicating that the first two components explain most of the variance.

2c

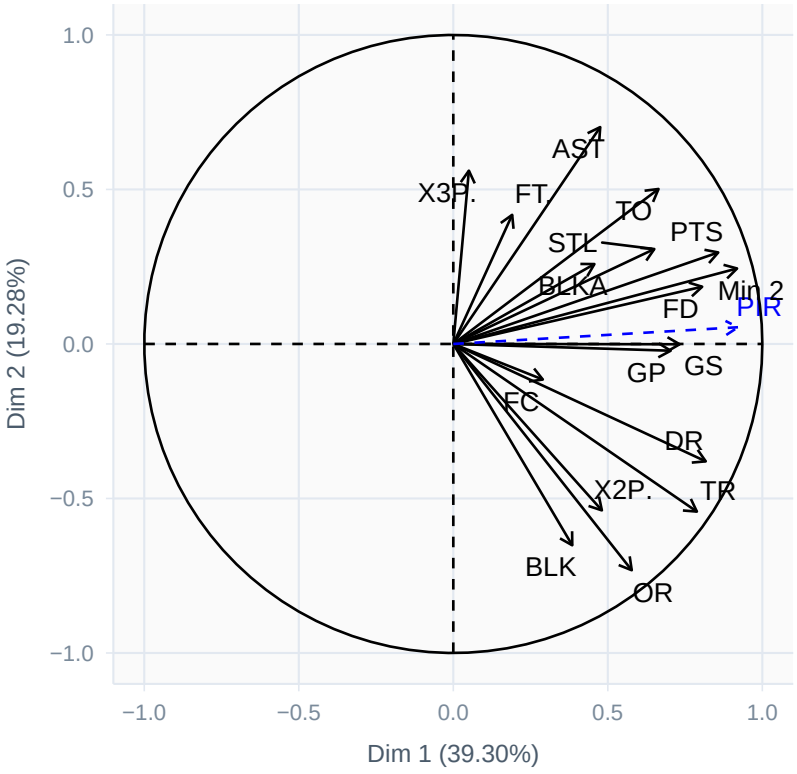
```
##            Dim.1         Dim.2       Dim.3        Dim.4        Dim.5
## GP    0.70345944 -0.0218546402  0.46662016 -0.056788043  0.219930804
## GS    0.73246819 -0.0009066644  0.23064869 -0.078071104  0.159535929
## PTS   0.85736119  0.2960999943 -0.02204878 -0.157100124 -0.018189268
## X2P.  0.48077204 -0.5384501095  0.18734335  0.245175958 -0.261173909
## X3P.  0.05039609  0.5607721596  0.63038821  0.070943172  0.250469683
## FT.   0.19057912  0.4178613912 -0.08333394 -0.578772875 -0.121321703
## OR    0.57741509 -0.7321762370 -0.06182125 -0.100781973 -0.037612249
## DR    0.81696083 -0.3798924520  0.12224829 -0.159523796  0.008739367
## TR    0.78827264 -0.5424339522  0.06342686 -0.145476847 -0.015638587
## AST   0.47544687  0.7017542494 -0.09120426  0.155127098 -0.256352578
## STL   0.65047380  0.3071519638  0.15536757  0.280322422 -0.327857982
## TO    0.66412520  0.5014585478 -0.36043627  0.205470148  0.029433507
## BLK   0.38495635 -0.6514179288 -0.24511167 -0.004224641  0.086962077
## BLKA  0.45663599  0.2585049001 -0.43397239 -0.211570921  0.533758547
```

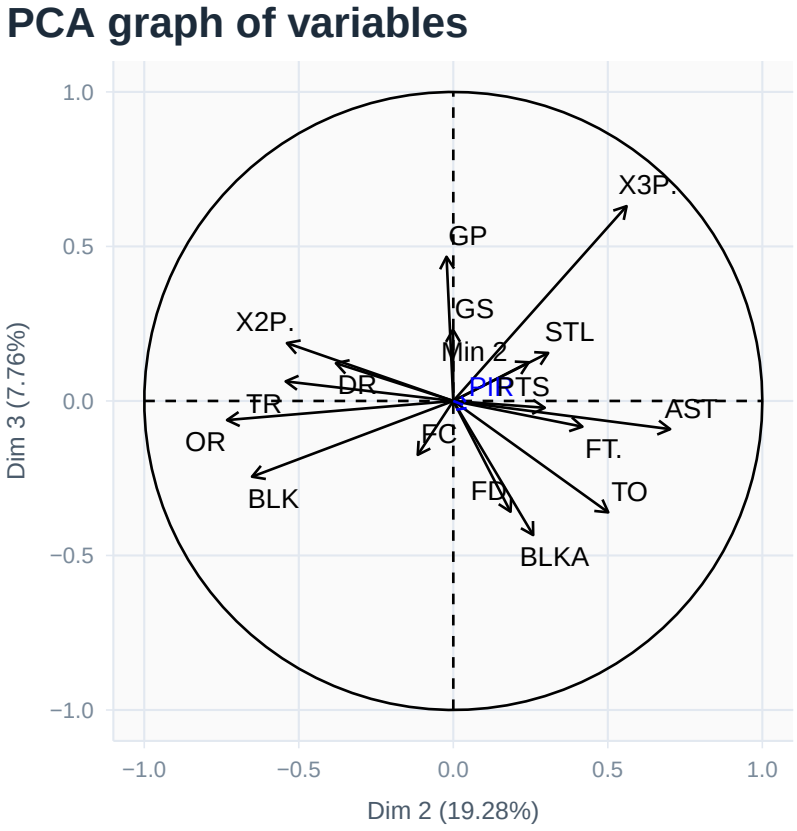
```
## FC    0.28967268 -0.1150539616 -0.17437917  0.735470315  0.313240362
## FD    0.80535676  0.1852181747 -0.35843977 -0.005248291 -0.222665236
## Min 2 0.91830049  0.2442284079  0.12519550  0.022578337  0.036994593
```

```
##            Dim.1         Dim.2       Dim.3        Dim.4        Dim.5
## GP    0.70345944 -0.0218546402  0.46662016 -0.056788043  0.219930804
## GS    0.73246819 -0.0009066644  0.23064869 -0.078071104  0.159535929
## PTS   0.85736119  0.2960999943 -0.02204878 -0.157100124 -0.018189268
## X2P.  0.48077204 -0.5384501095  0.18734335  0.245175958 -0.261173909
## X3P.  0.05039609  0.5607721596  0.63038821  0.070943172  0.250469683
## FT.   0.19057912  0.4178613912 -0.08333394 -0.578772875 -0.121321703
## OR    0.57741509 -0.7321762370 -0.06182125 -0.100781973 -0.037612249
## DR    0.81696083 -0.3798924520  0.12224829 -0.159523796  0.008739367
## TR    0.78827264 -0.5424339522  0.06342686 -0.145476847 -0.015638587
## AST   0.47544687  0.7017542494 -0.09120426  0.155127098 -0.256352578
## STL   0.65047380  0.3071519638  0.15536757  0.280322422 -0.327857982
## TO    0.66412520  0.5014585478 -0.36043627  0.205470148  0.029433507
## BLK   0.38495635 -0.6514179288 -0.24511167 -0.004224641  0.086962077
## BLKA  0.45663599  0.2585049001 -0.43397239 -0.211570921  0.533758547
## FC    0.28967268 -0.1150539616 -0.17437917  0.735470315  0.313240362
## FD    0.80535676  0.1852181747 -0.35843977 -0.005248291 -0.222665236
## Min 2 0.91830049  0.2442284079  0.12519550  0.022578337  0.036994593
```

```
##             Dim.1        Dim.2       Dim.3        Dim.4        Dim.5
## GP     7.40649988 1.456879e-02 16.50772644  0.263904523  5.777843398
## GS     8.02994263 2.507433e-05  4.03331600  0.498785325  3.040259518
## PTS   11.00176959 2.674318e+00  0.03685778  2.019696372  0.039520649
## X2P.   3.45949980 8.843562e+00  2.66095055  4.919135548  8.148041349
## X3P.   0.03801269 9.592000e+00 30.12842490  0.411864385  7.493832395
## FT.    0.54360764 5.325995e+00  0.52650689 27.412545597  1.758209566
## OR     4.99012194 1.635187e+01  0.28975829  0.831186999  0.168986591
## DR     9.98935324 4.402077e+00  1.13304152  2.082495053  0.009123333
## TR     9.30010403 8.974908e+00  0.30500480  1.731891923  0.029213842
## AST    3.38328741 1.502126e+01  0.63065300  1.969284068  7.849988576
## STL    6.33278166 2.877682e+00  1.83012602  6.430558626 12.840004266
## TO     6.60138142 7.670194e+00  9.84956897  3.454860337  0.103485066
## BLK    2.21798123 1.294362e+01  4.55499857  0.001460537  0.903344602
## BLKA   3.12086634 2.038327e+00 14.27856190  3.663068026 34.031675339
## FC     1.25588532 4.037750e-01  2.30541695 44.265291234 11.720579315
## FD     9.70759217 1.046414e+00  9.74075572  0.002254077  5.922410119
## Min 2 12.62131301 1.819402e+00  1.18833170  0.041717370  0.163482076
```

PCA graph of variables





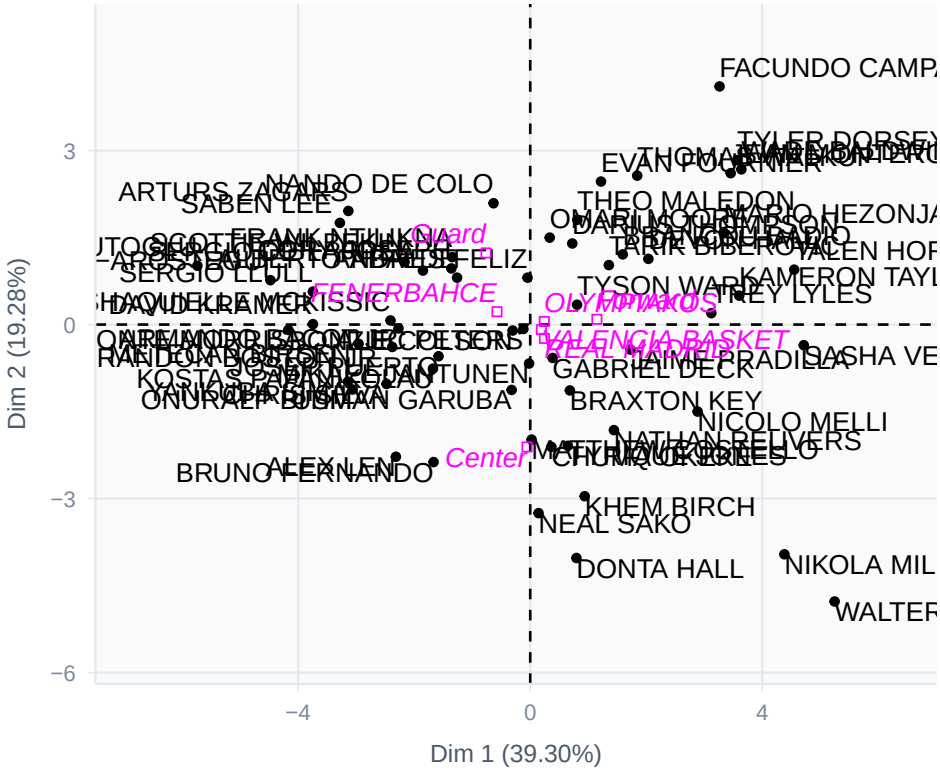
2d

PC1 (39.30%) – Overall Player Activity: Most statistical variables are positively correlated with PC1, particularly points, minutes played, and rebounds. This dimension represents overall player activity and statistical contribution.

PC2 (19.28%) – Playmaking vs. Rebounding: Assists, turnovers, and steals correlate positively with PC2, while rebounds and blocks correlate negatively. This dimension distinguishes playmaking-oriented players from inside players.

PC3 (7.76%) – Three-Point Shooting vs. Physical Play: Three-point shooting percentage and games played correlate positively with PC3, while blocks against and turnovers correlate negatively. This dimension may reflect differences in playing style, although its interpretation is less clear.

PCA graph of individuals



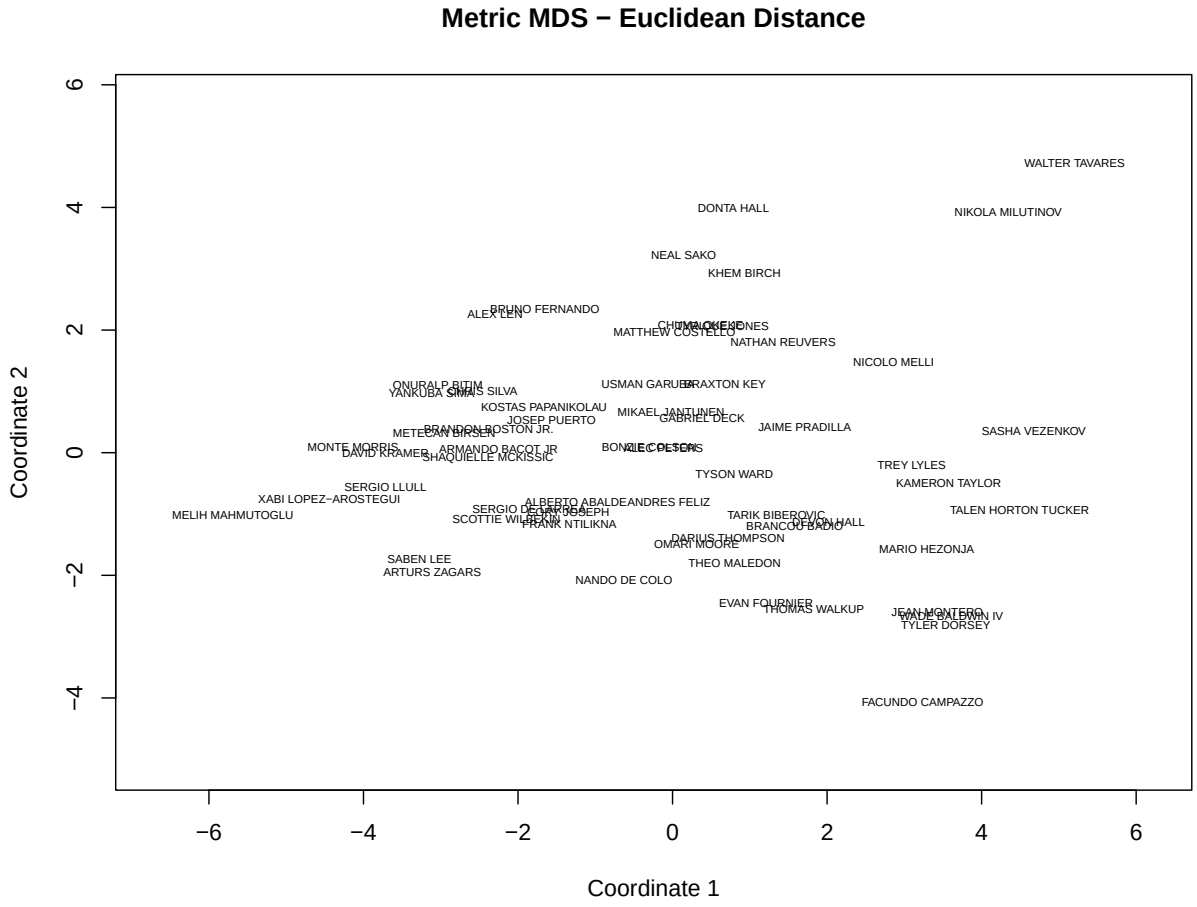


The PC1–PC2 plot (58.58% explained variance) separates players mainly by overall statistical activity (PC1) and playing style (PC2). Players 33 and 14 show high PC1 scores, while player 44 has a low score. PC2 distinguishes playmaking-oriented players, such as player 24, from more inside-oriented players, such as players 33 and 17.

The PC2–PC3 plot (27.04% explained variance) highlights additional differences in playing style, particularly three-point shooting. Players 25 and 23 have relatively high PC3 scores. Player positions partly reflect their roles, with guards tending towards higher PC2 values and centers towards lower values. However, the teams overlap considerably, indicating no clear separation by team.

3. Multidimensional Scaling (MDS)

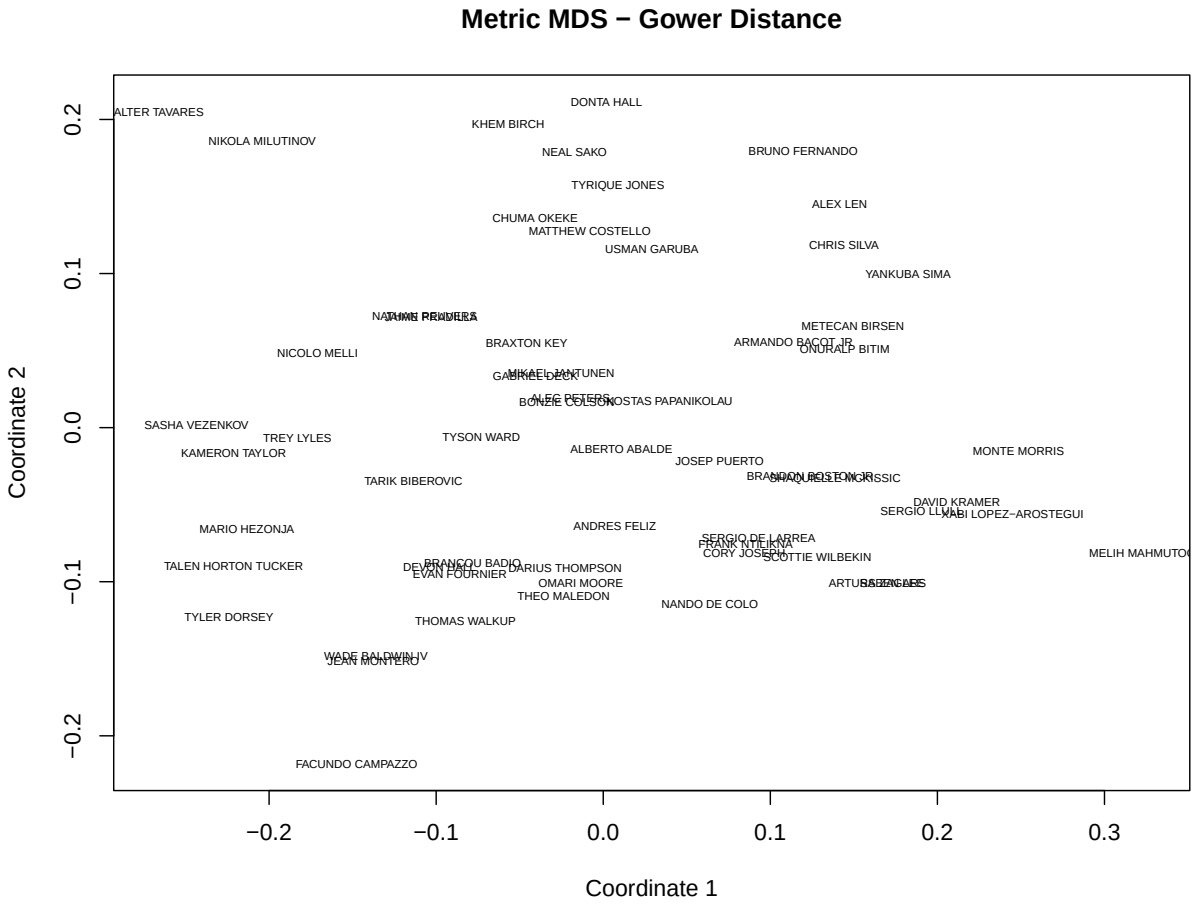
3a. Metric MDS - Euclidean Distance



3b

The Euclidean MDS plot shows a central concentration of players with relatively similar statistical profiles, without clearly separated clusters. Some players, such as Walter Tavares, Nikola Milutinov, and Facundo Campazzo, occupy more isolated positions, indicating distinct playing characteristics. Overall, the plot reveals both similarities among many players and substantial differences for certain individuals.

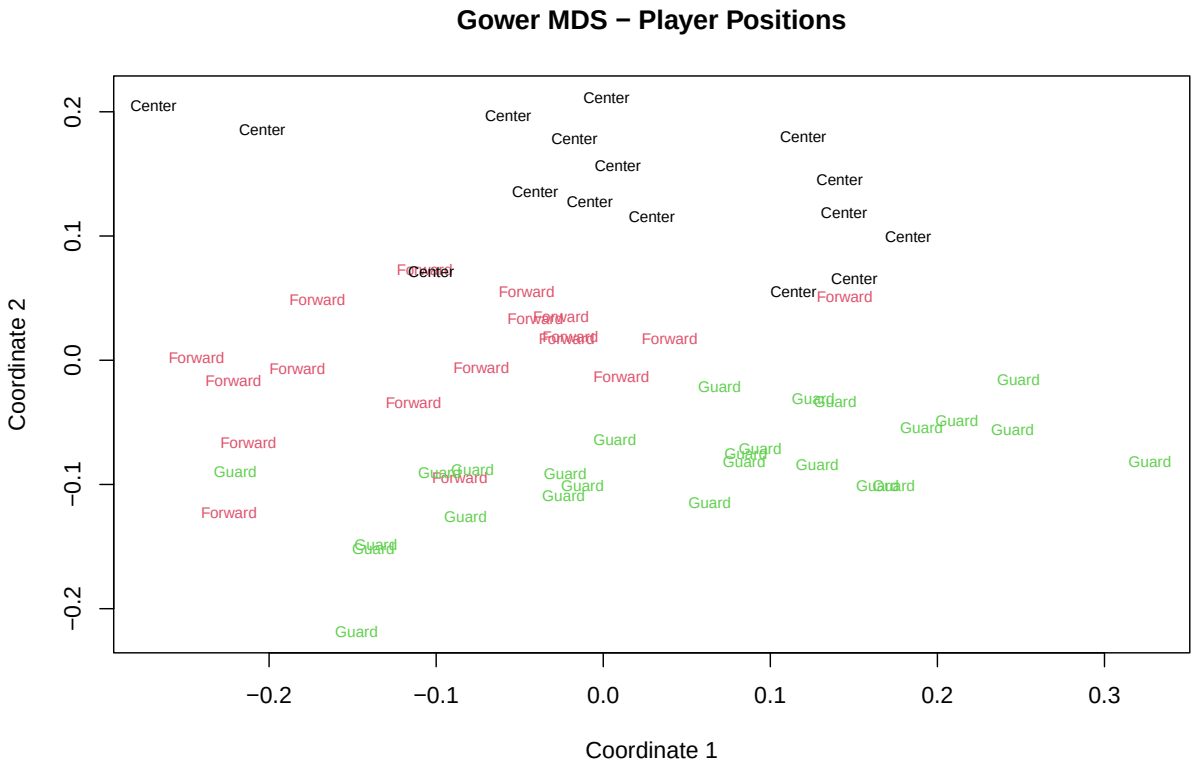
3c. Metric MDS - Gower Distance

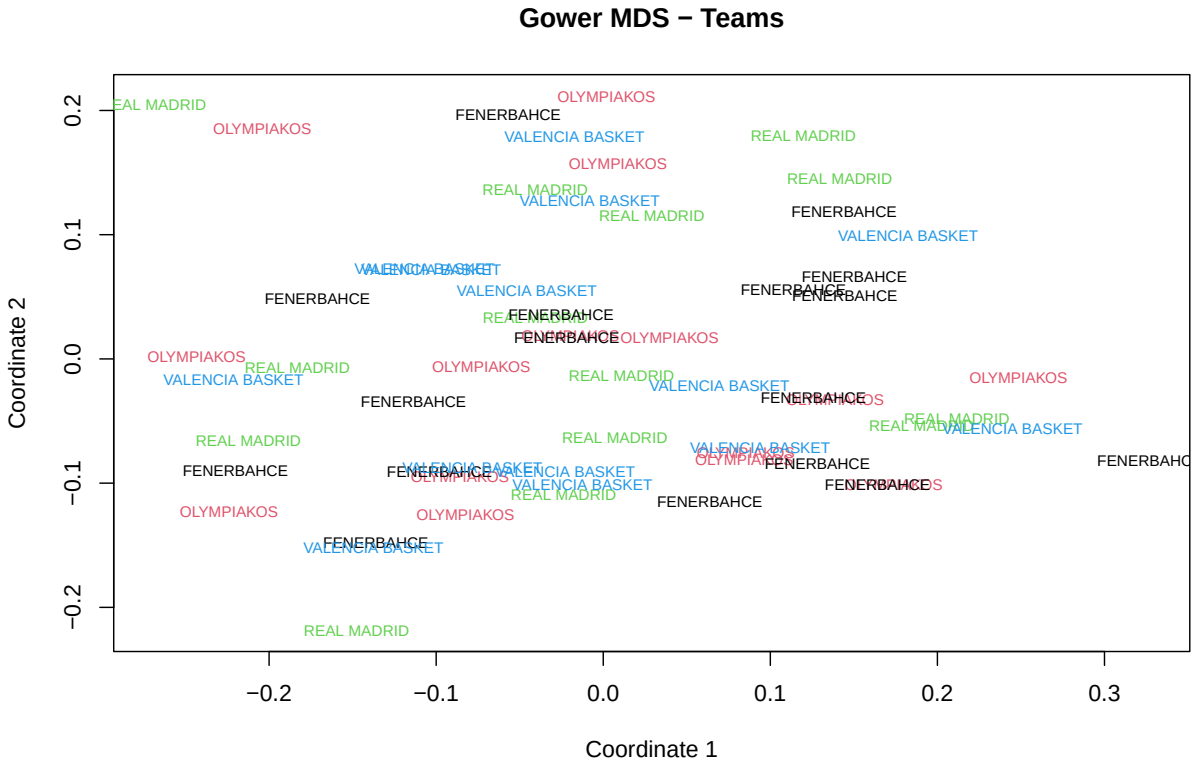


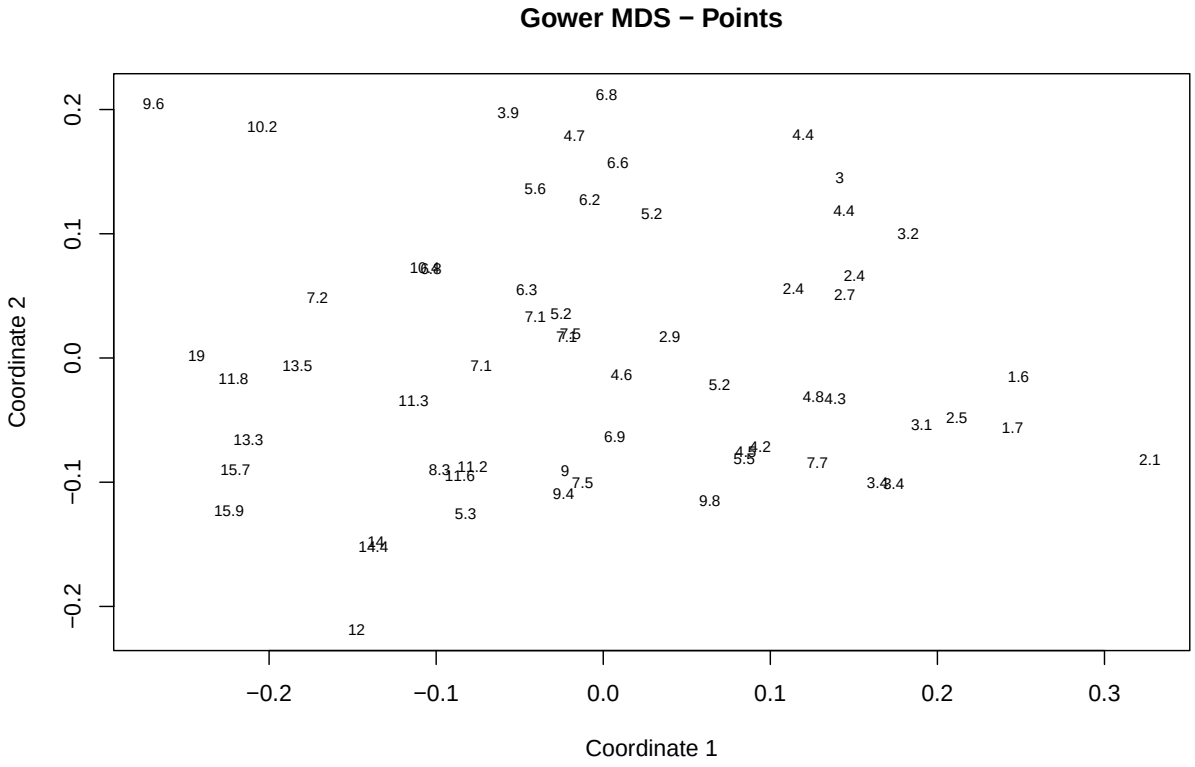

```
##  [1]  1.168329e+00  6.441960e-01  2.886618e-01  1.697698e-01  1.391959e-01
##  [6]  1.135799e-01  1.061880e-01  7.372461e-02  6.500444e-02  5.112237e-02
## [11]  4.733844e-02  4.145983e-02  3.451877e-02  3.114349e-02  2.811408e-02
## [16]  2.458142e-02  2.243850e-02  1.816294e-02  1.734874e-02  1.383134e-02
## [21]  1.329685e-02  8.292776e-03  7.511550e-03  5.477563e-03  4.965205e-03
## [26]  3.008304e-03  2.734736e-03  9.139220e-04 -8.326673e-17 -1.985155e-04
## [31] -7.636607e-04 -1.439072e-03 -2.367760e-03 -4.290621e-03 -4.528827e-03
## [36] -5.504499e-03 -6.442520e-03 -7.685371e-03 -8.305791e-03 -8.546051e-03
## [41] -9.790315e-03 -1.014870e-02 -1.073121e-02 -1.134592e-02 -1.258196e-02
## [46] -1.458524e-02 -1.549999e-02 -1.555963e-02 -1.675239e-02 -1.844129e-02
## [51] -1.952001e-02 -2.235721e-02 -2.620667e-02 -2.666931e-02 -3.098803e-02
## [56] -3.587956e-02 -3.730000e-02 -4.059799e-02 -5.252564e-02 -6.568356e-02
```

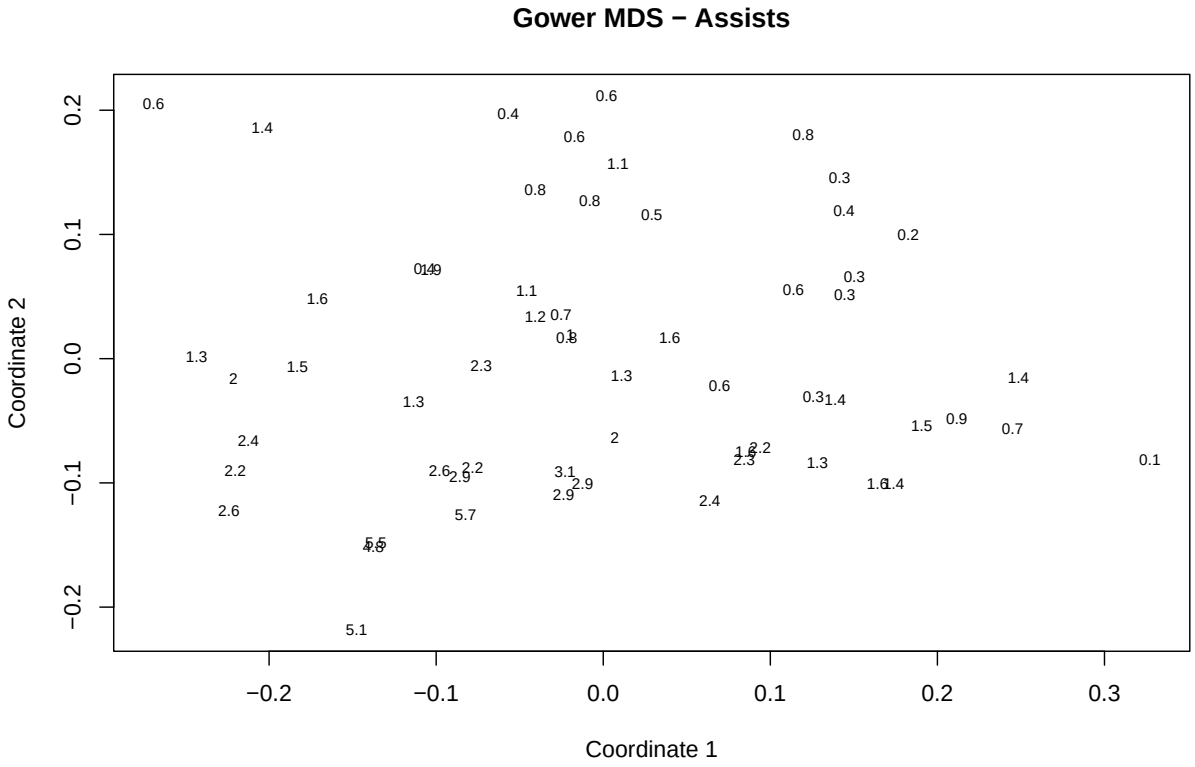
```
## [1] 0.4914459 0.5763360
```

3d. Interpretation of Gower MDS Dimensions









Coordinate 1 – Scoring Activity: Players with higher average points generally appear on the left, while

lower-scoring players tend to appear on the right, suggesting that this dimension partly reflects scoring activity.

Coordinate 2 – Playing Position and Playmaking: Centers are mainly located at positive values, guards at negative values, and forwards in between. Higher assist averages also tend to occur at negative values, indicating that this dimension distinguishes inside-oriented players from playmakers.

Team Membership: Players from all four teams overlap considerably, suggesting that team membership does not strongly explain the MDS dimensions.

3e

Gower MDS provides clearer grouping of players by position, with centers, guards, and forwards showing a more distinct separation. This is because Gower distance incorporates the categorical variable POSITION alongside numerical statistics. Euclidean MDS mainly reflects differences in numerical performance and shows less distinct groups. # 4. Multiple Correspondence Analysis (MCA)

4a. Convert Numerical Variables to Binary Categories

```
## 'data.frame':    60 obs. of  20 variables:
##  $ TEAM    : Factor w/ 4 levels "FENERBAHCE","OLYMPIAKOS",..: 2 2 2 2 2 2 2 2 2 2 ...
##  $ POSITION: Factor w/ 3 levels "Center","Forward",..: 3 3 2 3 3 2 2 2 2 1 ...
##  $ GP      : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 2 2 2 2 ...
##  $ GS      : Factor w/ 2 levels "Below average",..: 2 1 1 1 1 2 2 2 1 2 ...
##  $ PTS     : Factor w/ 2 levels "Below average",..: 1 1 1 1 1 2 1 2 2 2 ...
##  $ X2P.    : Factor w/ 2 levels "Below average",..: 1 1 2 1 1 2 2 1 2 2 ...
##  $ X3P.    : Factor w/ 2 levels "Below average",..: 2 2 1 2 1 2 2 2 2 1 ...
##  $ FT.     : Factor w/ 2 levels "Below average",..: 2 2 1 2 1 2 1 2 2 2 ...
##  $ OR      : Factor w/ 2 levels "Below average",..: 1 1 1 1 1 2 1 1 1 2 ...
##  $ DR      : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 1 1 2 2 ...
##  $ TR      : Factor w/ 2 levels "Below average",..: 1 1 2 1 1 2 1 1 2 2 ...
##  $ AST     : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 1 1 2 1 1 ...
##  $ STL     : Factor w/ 2 levels "Below average",..: 2 1 1 1 1 2 1 2 1 2 ...
##  $ TO      : Factor w/ 2 levels "Below average",..: 2 2 2 2 1 2 1 2 1 1 ...
##  $ BLK     : Factor w/ 2 levels "Below average",..: 1 2 1 1 1 1 1 1 1 2 ...
##  $ BLKA    : Factor w/ 2 levels "Below average",..: 1 2 2 2 2 2 1 2 1 2 ...
##  $ FC      : Factor w/ 2 levels "Below average",..: 2 2 2 1 1 1 1 2 2 1 ...
##  $ FD      : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 1 2 2 2 ...
##  $ PIR     : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 1 2 2 2 ...
##  $ Min 2   : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 1 2 1 2 ...
```

```
##               TEAM       POSITION              GP                  GS
##  FENERBAHCE     :17   Center :16   Below average:21   Below average:37
##  OLYMPIAKOS     :15   Forward:18   Over average :39   Over average :23
##  REAL MADRID    :14   Guard  :26
##  VALENCIA BASKET:14
##             PTS                X2P.               X3P.                FT.
##  Below average:37   Below average:30   Below average:19   Below average:25
##  Over average :23   Over average :30   Over average :41   Over average :35
##
##
##              OR                 DR                 TR                 AST
```

```
##  Below average:36   Below average:33   Below average:32   Below average:40
##  Over average :24   Over average :27   Over average :28   Over average :20
##
##
##             STL                TO                BLK                BLKA
##  Below average:35   Below average:34   Below average:38   Below average:26
##  Over average :25   Over average :26   Over average :22   Over average :34
##
##
##              FC                FD                PIR               Min 2
##  Below average:28   Below average:33   Below average:31   Below average:32
##  Over average :32   Over average :27   Over average :29   Over average :28
##
##
```

4b. Multiple Correspondence Analysis

```
##
## Call:
## MCA(X = mca_data, graph = FALSE)
##
##
## Eigenvalues
##                        Dim.1  Dim.2  Dim.3  Dim.4  Dim.5
## Variance               0.293  0.191  0.079  0.074  0.067
## % of var.             25.463 16.618  6.883  6.434  5.824
## Cumulative % of var. 25.463 42.081 48.964 55.398 61.222
##
## Individuals (the 10 first)
##                          Dim.1    ctr   cos2    Dim.2    ctr   cos2    Dim.3
## THOMAS WALKUP         |  0.423  1.020  0.156 | -0.511  2.276  0.228 | -0.126
## FRANK NTILIKNA        | -0.490  1.365  0.234 | -0.309  0.835  0.094 | -0.100
## TYSON WARD            |  0.433  1.068  0.154 |  0.014  0.002  0.000 | -0.260
## SABEN LEE             | -0.605  2.086  0.374 | -0.460  1.847  0.216 | -0.100
## MONTE MORRIS          | -0.763  3.314  0.544 | -0.138  0.167  0.018 |  0.010
## SASHA VEZENKOV        |  0.777  3.434  0.497 |  0.013  0.002  0.000 | -0.275
## KOSTAS PAPANIKOLAU    | -0.420  1.004  0.168 |  0.063  0.034  0.004 | -0.052
## TYLER DORSEY          |  0.572  1.865  0.273 | -0.694  4.203  0.402 | -0.118
## ALEC PETERS           |  0.207  0.243  0.040 |  0.097  0.082  0.009 | -0.640
## NIKOLA MILUTINOV      |  0.658  2.464  0.321 |  0.486  2.058  0.175 | -0.167
##                          ctr   cos2
## THOMAS WALKUP          0.334  0.014 |
## FRANK NTILIKNA         0.210  0.010 |
## TYSON WARD             1.418  0.055 |
## SABEN LEE              0.209  0.010 |
## MONTE MORRIS           0.002  0.000 |
## SASHA VEZENKOV         1.595  0.062 |
## KOSTAS PAPANIKOLAU     0.058  0.003 |
## TYLER DORSEY           0.294  0.012 |
## ALEC PETERS            8.611  0.386 |
## NIKOLA MILUTINOV       0.584  0.021 |
##
## Categories (the 10 first)
```

```
##                          Dim.1    ctr   cos2 v.test    Dim.2    ctr   cos2
## FENERBAHCE             | -0.202  0.197  0.016 -0.974 | -0.081  0.049  0.003
## OLYMPIAKOS             |  0.020  0.002  0.000  0.088 | -0.159  0.164  0.008
## REAL MADRID            |  0.028  0.003  0.000  0.118 |  0.172  0.181  0.009
## VALENCIA BASKET        |  0.196  0.153  0.012  0.829 |  0.097  0.057  0.003
## Center                 | -0.181  0.148  0.012 -0.836 |  1.127  8.856  0.462
## Forward                |  0.548  1.540  0.129  2.757 |  0.073  0.042  0.002
## Guard                  | -0.268  0.533  0.055 -1.803 | -0.744  6.275  0.423
## GP_Below average       | -1.004  6.019  0.542 -5.656 | -0.175  0.280  0.016
## GP_Over average        |  0.540  3.241  0.542  5.656 |  0.094  0.151  0.016
## GS_Below average       | -0.527  2.924  0.447 -5.134 |  0.024  0.009  0.001
##                        v.test    Dim.3    ctr   cos2 v.test
## FENERBAHCE             -0.393 | -0.834 12.463  0.275 -4.030 |
## OLYMPIAKOS             -0.703 | -0.688  7.466  0.158 -3.049 |
## REAL MADRID             0.729 |  0.561  4.639  0.096  2.377 |
## VALENCIA BASKET         0.409 |  1.189 20.837  0.430  5.039 |
## Center                  5.219 |  0.254  1.084  0.023  1.175 |
## Forward                 0.368 | -0.545  5.625  0.127 -2.740 |
## Guard                  -4.997 |  0.221  1.337  0.037  1.485 |
## GP_Below average       -0.986 | -0.230  1.172  0.029 -1.298 |
## GP_Over average         0.986 |  0.124  0.631  0.029  1.298 |
## GS_Below average        0.232 | -0.185  1.332  0.055 -1.802 |
##
## Categorical variables (eta2)
##                          Dim.1 Dim.2 Dim.3
## TEAM                   | 0.021 0.017 0.719 |
## POSITION               | 0.130 0.580 0.127 |
## GP                     | 0.542 0.016 0.029 |
## GS                     | 0.447 0.001 0.055 |
## PTS                    | 0.564 0.134 0.004 |
## X2P.                   | 0.047 0.287 0.005 |
## X3P.                   | 0.002 0.161 0.017 |
## FT.                    | 0.042 0.215 0.224 |
## OR                     | 0.128 0.555 0.026 |
## DR                     | 0.423 0.323 0.035 |

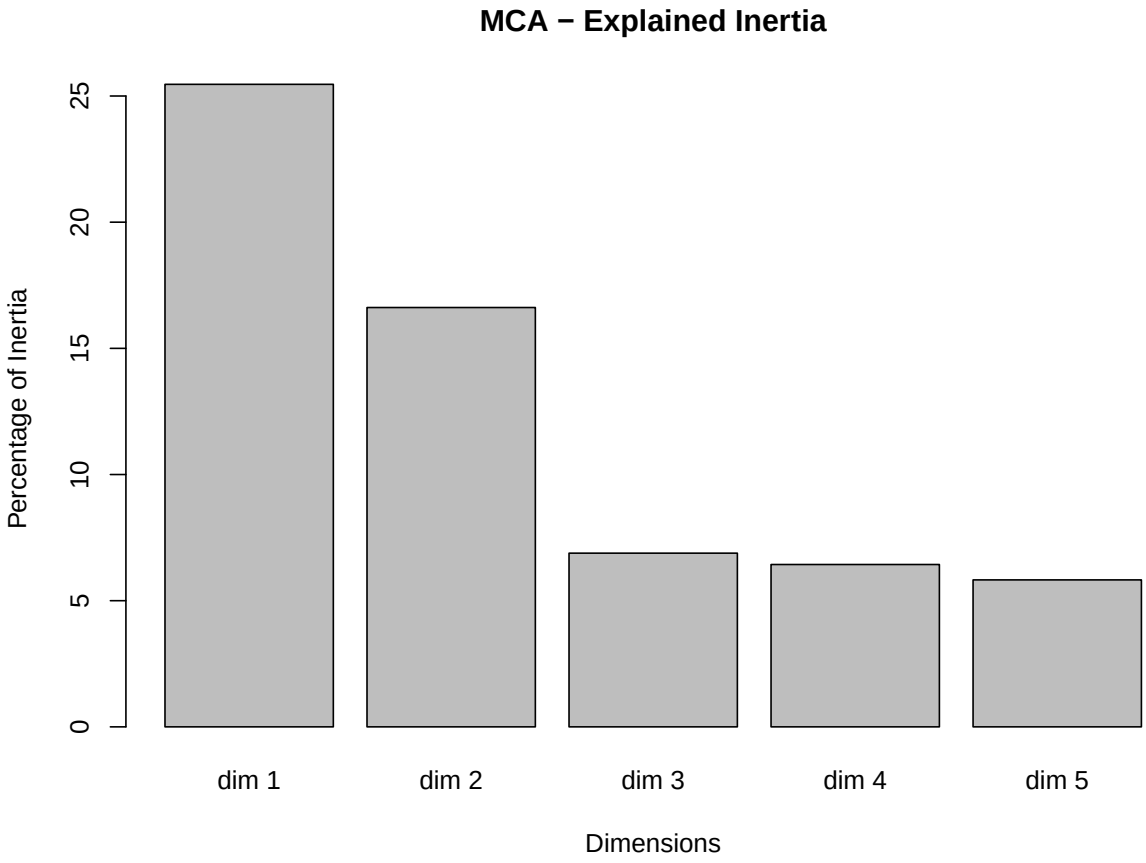
##       eigenvalue percentage of variance cumulative percentage of variance
## dim 1 0.29282430              25.462983                           25.46298
## dim 2 0.19110749              16.618043                           42.08103
## dim 3 0.07915964               6.883447                           48.96447
## dim 4 0.07398656               6.433614                           55.39809
## dim 5 0.06697365               5.823796                           61.22188

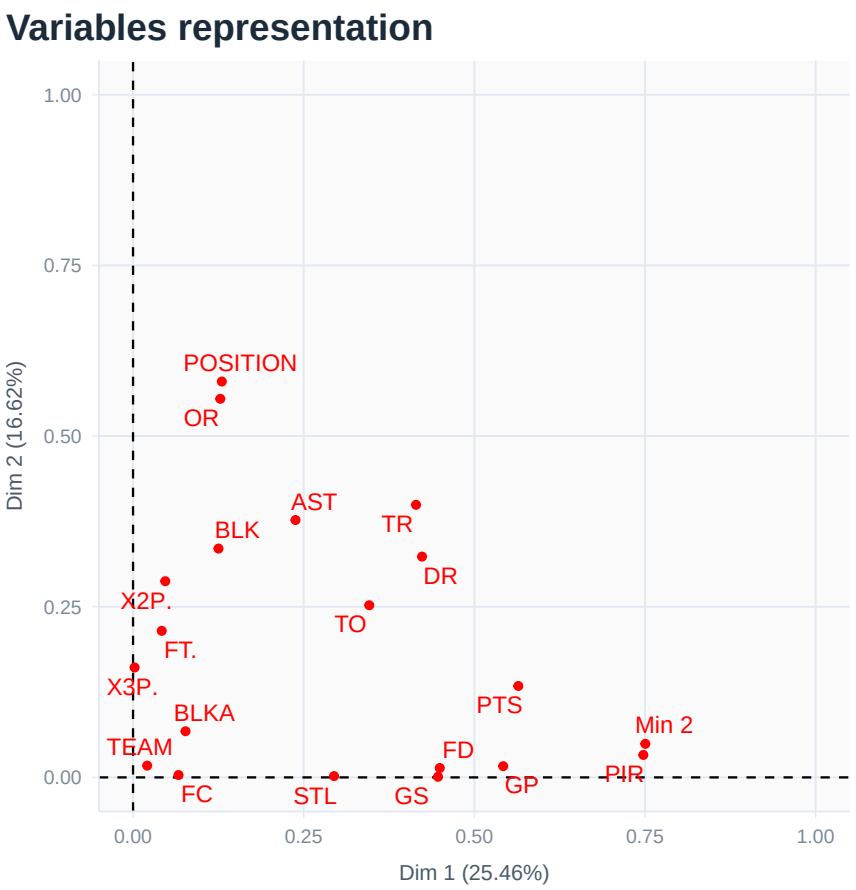
## $`Dim 1`
##
## Link between the variable and the categorical variable (1-way anova)
## ============================================
##                  R2      p.value
## Min 2    0.75049087 3.924760e-19
## PIR      0.74759117 5.497210e-19
## PTS      0.56439705 4.685675e-12
## GP       0.54230206 2.004974e-11
## FD       0.44935765 4.664512e-09
## GS       0.44674418 5.365644e-09
```

```
## DR       0.42338548 1.825151e-08
## TR       0.41456907 2.861885e-08
## TO       0.34606063 7.701888e-07
## STL      0.29445518 7.501258e-06
## AST      0.23804330 7.678147e-05
## OR       0.12787760 5.030215e-03
## BLK      0.12521652 5.542554e-03
## POSITION 0.13009925 1.883157e-02
## BLKA     0.07690254 3.194181e-02
## FC       0.06666730 4.638722e-02
##
## Link between variable and the categories of the categorical variables
## ============================================================
##                              Estimate      p.value
## Min 2=Min 2_Over average    0.4698330 3.924760e-19
## PIR=PIR_Over average        0.4681414 5.497210e-19
## PTS=PTS_Over average        0.4180733 4.685675e-12
## GP=GP_Over average          0.4177375 2.004974e-11
## FD=FD_Over average          0.3645709 4.664512e-09
## GS=GS_Over average          0.3719542 5.365644e-09
## DR=DR_Over average          0.3538783 1.825151e-08
## TR=TR_Over average          0.3491960 2.861885e-08
## TO=TO_Over average          0.3211994 7.701888e-07
## STL=STL_Over average        0.2978039 7.501258e-06
## AST=AST_Over average        0.2800321 7.678147e-05
## POSITION=Forward            0.2787756 4.859760e-03
## OR=OR_Over average          0.1974991 5.030215e-03
## BLK=BLK_Over average        0.1986792 5.542554e-03
## BLKA=BLKA_Over average      0.1514150 3.194181e-02
## FC=FC_Over average          0.1400320 4.638722e-02
## FC=FC_Below average        -0.1400320 4.638722e-02
## BLKA=BLKA_Below average    -0.1514150 3.194181e-02
## BLK=BLK_Below average      -0.1986792 5.542554e-03
## OR=OR_Below average        -0.1974991 5.030215e-03
## AST=AST_Below average      -0.2800321 7.678147e-05
## STL=STL_Below average      -0.2978039 7.501258e-06
## TO=TO_Below average        -0.3211994 7.701888e-07
## TR=TR_Below average        -0.3491960 2.861885e-08
## DR=DR_Below average        -0.3538783 1.825151e-08
## GS=GS_Below average        -0.3719542 5.365644e-09
## FD=FD_Below average        -0.3645709 4.664512e-09
## GP=GP_Below average        -0.4177375 2.004974e-11
## PTS=PTS_Below average      -0.4180733 4.685675e-12
## PIR=PIR_Below average      -0.4681414 5.497210e-19
## Min 2=Min 2_Below average -0.4698330 3.924760e-19
##
## $`Dim 2`
##
## Link between the variable and the categorical variable (1-way anova)
## ============================================
##                  R2      p.value
## OR       0.55460719 9.000731e-12
## POSITION 0.57992609 1.839852e-11
## TR       0.39928150 6.149994e-08
```

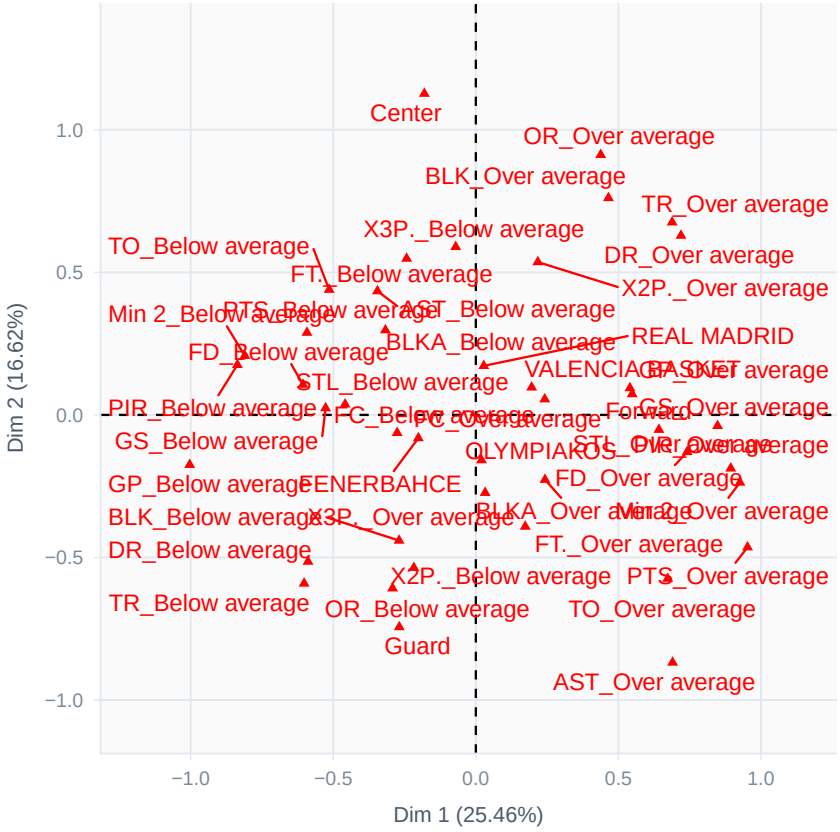
```
## AST      0.37703318 1.812981e-07
## BLK      0.33520399 1.259809e-06
## DR       0.32344402 2.129237e-06
## X2P.     0.28739466 1.012286e-05
## TO       0.25219391 4.345197e-05
## FT.      0.21469340 1.928376e-04
## X3P.     0.16097706 1.487605e-03
## PTS      0.13389107 4.038357e-03
## BLKA     0.06759126 4.484562e-02
##
## Link between variable and the categories of the categorical variables
## ============================================================
##                             Estimate      p.value
## OR=OR_Over average       0.3322737 9.000731e-12
## POSITION=Center          0.4260985 2.403476e-09
## TR=TR_Over average       0.2768507 6.149994e-08
## AST=AST_Below average    0.2847114 1.812981e-07
## BLK=BLK_Over average     0.2626102 1.259809e-06
## DR=DR_Over average       0.2498739 2.129237e-06
## X2P.=X2P._Over average   0.2343571 1.012286e-05
## TO=TO_Below average      0.2215140 4.345197e-05
## FT.=FT._Below average    0.2054307 1.928376e-04
## X3P.=X3P._Below average  0.1885269 1.487605e-03
## PTS=PTS_Below average    0.1645020 4.038357e-03
## BLKA=BLKA_Below average  0.1146778 4.484562e-02
## BLKA=BLKA_Over average  -0.1146778 4.484562e-02
## PTS=PTS_Over average    -0.1645020 4.038357e-03
## X3P.=X3P._Over average  -0.1885269 1.487605e-03
## FT.=FT._Over average    -0.2054307 1.928376e-04
## TO=TO_Over average      -0.2215140 4.345197e-05
## X2P.=X2P._Below average -0.2343571 1.012286e-05
## DR=DR_Below average     -0.2498739 2.129237e-06
## BLK=BLK_Below average   -0.2626102 1.259809e-06
## AST=AST_Over average    -0.2847114 1.812981e-07
## TR=TR_Below average     -0.2768507 6.149994e-08
## POSITION=Guard          -0.3916480 1.839051e-08
## OR=OR_Below average     -0.3322737 9.000731e-12
##
## $`Dim 3`
##
## Link between the variable and the categorical variable (1-way anova)
## ============================================
##                  R2      p.value
## TEAM     0.71884776 1.920278e-15
## FT.      0.22362755 1.359259e-04
## FD       0.12369051 5.859276e-03
## POSITION 0.12738633 2.057915e-02
## BLKA     0.07778667 3.093214e-02
## AST      0.06905669 4.250729e-02
##
## Link between variable and the categories of the categorical variables
## ============================================================
##                               Estimate      p.value
## TEAM=VALENCIA BASKET     0.31850600 1.277497e-08
```

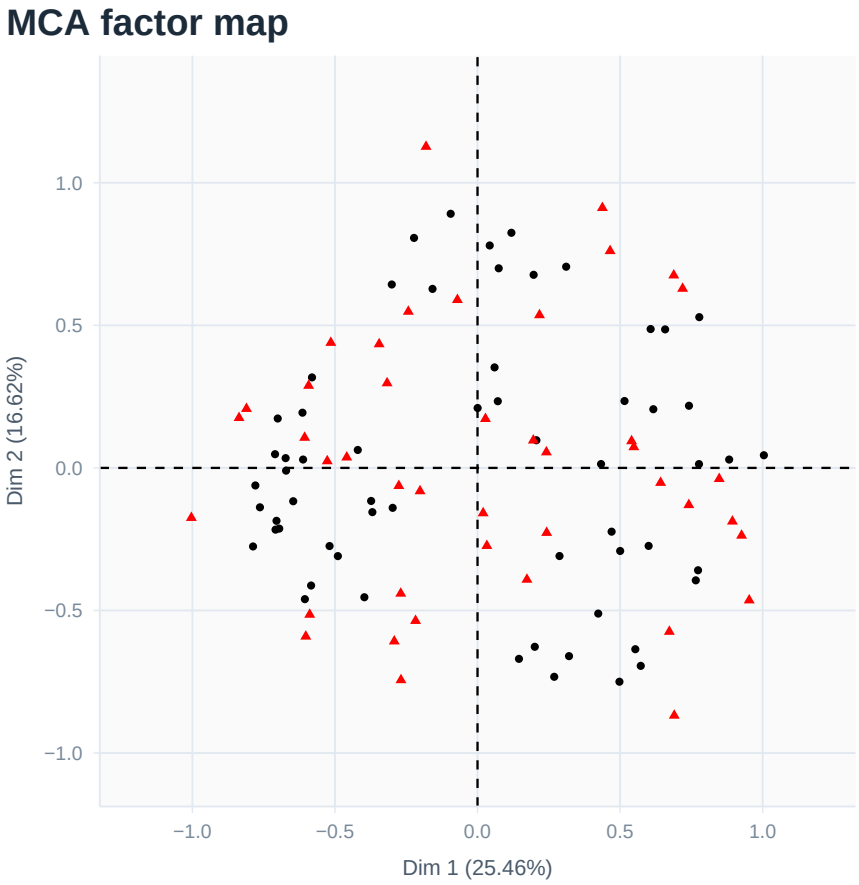
```
## FT.=FT._Below average    0.13493722 1.359259e-04
## FD=FD_Below average      0.09944947 5.859276e-03
## TEAM=REAL MADRID         0.14181455 1.611014e-02
## BLKA=BLKA_Over average   0.07917710 3.093214e-02
## AST=AST_Over average     0.07842076 4.250729e-02
## AST=AST_Below average   -0.07842076 4.250729e-02
## BLKA=BLKA_Below average -0.07917710 3.093214e-02
## FD=FD_Over average      -0.09944947 5.859276e-03
## POSITION=Forward        -0.14671615 5.154202e-03
## TEAM=OLYMPIAKOS         -0.20949884 1.685833e-03
## FT.=FT._Over average    -0.13493722 1.359259e-04
## TEAM=FENERBAHCE         -0.25082170 1.680275e-05
```




Variables representation
1.00
0.75
0.50
0.25
0.00
Dim 2 (16.62%)
POSITION
OR
AST
BLK
TR
DR
X2P.
TO
FT.
X3P.
PTS
BLKA
Min 2
TEAM
FD
PIR
FC
STL
GS
GP
0.00
0.25
0.50
0.75
1.00
Dim 1 (25.46%)

MCA − Categories





The first two MCA dimensions explain **42.08% of the total inertia** (25.46% and 16.62%).

Dimension 1 – Overall Statistical Involvement: This dimension is mainly associated with minutes played ($R^2 = 0.750$), PIR ($R^2 = 0.748$), and points scored ($R^2 = 0.564$). Above-average performance categories are located on the positive side, while below-average categories appear on the negative side.

Dimension 2 – Inside Play vs. Playmaking: Playing position ($R^2 = 0.580$), offensive rebounds ($R^2 = 0.555$), and assists ($R^2 = 0.377$) strongly characterize this dimension. Centers and above-average rebounding categories are located at positive values, while guards and above-average assists appear at negative values.

The individual factor map shows differences in player profiles without clearly separated clusters. Team categories are located near the origin, suggesting little influence on the first two dimensions. However, Dimension 3 (6.88%) is strongly associated with team membership ($R^2 = 0.719$).

4c

All three methods reveal similar patterns. PCA and MCA identify overall statistical involvement as the first dimension and differences between playmaking and inside-oriented players as the second. Euclidean MDS highlights differences in numerical player statistics, while Gower MDS provides clearer separation by playing position. Unlike PCA, MCA uses categorized performance variables, resulting in some information loss but making above- and below-average performance profiles easier to interpret.